

Leonardo Kulon

Wilton High School Class of 2027

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EXPERIENCE

Stock Market Simulation

- Developed a paper trade simulation on multiple markets similar to [Investopedia Stock Simulator](#) for backtesting trade algorithms and multiplayer trading with friends.
- Utilized the Finnhub and Alpaca API for live/historical data on stocks and cryptocurrencies.
- Implemented technical indicators (Stochastic, Bull & Bear, RedK Everex) in [Rust](#), chosen for its performance characteristics in financial computation.

Robotics Catalyst Foundation - Founder & Executive Director

- Established and incorporated a 501(c)(3) nonprofit organization dedicated to increasing STEM participation and accessibility through robotics mentorship and resources.
- **Raised over \$14,000** in funding for robotics teams.
- [Provided fiscal sponsorship for local teams](#), engineering an automated donation portal utilizing **Typescript and SQL** to streamline their fundraising process.
- Designed and deployed a [centralized geospatial database](#)(HTML, Typescript, Python, SQL) that aggregates data across multiple robotics ecosystems enabling prospective students and mentors to connect with local robotics programs.

FTC Robotics Team #7034 - Captain

- Lead technical development for an award winning FIRST Tech Challenge team, driving the development of robots that **secured qualification for the [2025 World Championship](#) and a 2026 Premier event.**
- Optimized scoring efficiency and minimized human driver error as the sole programmer, utilizing **Java and OpenCV** to create advanced sensor-driven automation.
- Author and maintain a suite of open-source libraries, used by 250+ teams worldwide. This includes PID control libraries and sensor/control wrappers in Java.

Linux Server Management

- Provision and manage cloud infrastructure utilizing IONOS VPS hosting to deploy web applications, REST API servers, and backend databases on **Linux-based servers**. (NGINX, Docker, SSH Tunneling)
- Built and configured a desktop PC used as a TrueNAS server for: storing backup data, **running local LLM models with Ollama**, off-loaded neural network training, and as a development server for other projects.
 - AMD Ryzen 7 3700X
 - NVIDIA RTX 4060ti
 - 32GiB DDR4 RAM
- Streamlined application deployment on Linux VPS by utilizing **Docker for containerization/reproducibility and Git for version control.**

Melanoma Classification with Machine Learning

- Developed a **convolutional neural network** using Python and Tensorflow to autonomously classify skin lesions as benign or malignant, training and validating the classification model on a dataset of 10,000 medical images.
- Awarded 1st Place in Engineering in the Connecticut Science Fair.

Data Science Course

- Completed Kane's Data Science Course, a course taught by a former Harvard professor.
- Utilized **R & Git** to process and visualize a variety of data sources.

Wilton High School Robotics Club - Founder & President

- Started the school's first robotics club, growing membership to compete in the VEX competition.
- Taught new-to-STEM members fundamental concepts in CAD(**OnShape**), robot assembly, and programming(C++).
- Assisted in hosting the VEX State Championship hosted at the high school.

Sea Scouts Ship #84 - Leader

- Serve as the elected youth leader of Ship #84, directing a crew of 7 Scouts in sailing operations, seamanship, and maritime safety.
- Responsible for crew coordination, watch assignments, and upholding BSA Sea Scout standards.
- Certified American Red Cross Lifeguard.

EDUCATION

Completed Courses

- AP Computer Science A
- AP Chemistry
- AP English Language
- AP United States History
- Honors Precalculus 3
- Honors Algebra 2-3

Future Courses

- AP Physics
- AP Literature
- AP Calculus
- AP Economics

Standardized Testing

- SAT - 1500(730 English, 770 Math)